

DL3004 Final Project

Phase 3 - OLAP Analytics, Performance Optimization & BI Dashboard

Objective

Students turn their warehouse into a decision-support system.

Aligned Labs

- Lab 1: Analytical SQL
- Lab 4: OLAP
- Lab 9-12 concepts

1. Analytical SQL Layer

Students create queries using:

Aggregations:

- SUM
- AVG
- COUNT

Example:

Monthly Revenue

Month | Revenue | Growth

OLAP Queries

Must include:

ROLLUP

Example hierarchy:

Region

↓

Category

↓

Product

Output:

- Product Revenue
- Category Total
- Region Total
- Grand Total

CUBE

Example:

Region × Channel

2. Create Reporting Views

Students create:

- vw_monthly_sales
- vw_customer_analysis
- vw_product_performance

3. Query Optimization

Students perform:

Before:

EXPLAIN ANALYZE

Create indexes:

CREATE INDEX

After:

EXPLAIN ANALYZE

Compare:

- Execution time
- Query cost

4. Power BI Dashboard

Connect Power BI to warehouse.

Dashboard requirements:

Page 1: Executive Overview

Contains:

- Revenue KPI
- Orders KPI
- Customers KPI
- Sales trend

Page 2: Detailed Analysis

Contains:

- Product analysis
- Regional analysis
- Customer segmentation

Page 3: Decision Page

Students answer:

"Based on the dashboard, what decision should management take?"

Example:

Increase marketing investment in Electronics category because revenue grew 35%.

Phase 3 Deliverables

- Analytical SQL Queries (SQL)
- OLAP Queries (SQL)
- Reporting Views (SQL)
- Indexing Report (PDF)
- Power BI Dashboard (PBIX)
- Dashboard Screenshots (PDF)
- Business Insights Report (PDF)
- Final Presentation (PPT)

Phase 3 Marks: 30%

Analytical SQL: 10

OLAP Implementation: 5

Dashboard: 10

Business Insights: 5

Total: 30

Final Project Overall Evaluation

Phase 1: Design - 30%

Phase 2: Implementation - 40%

Phase 3: Analytics & BI - 30%

Total: 100%